Varun Ganjigunte Prakash

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PROFILE

Computer vision and applied AI engineer with production experience in multimodal video understanding, object detection fine-tuning, document AI, and speech biometrics for clinical healthcare. Five peer-reviewed publications in computer vision for autism assessment and human action recognition.

EDUCATION

Bachelor of Engineering in Electronics and Communication Aug 2014 - May 2018
Sri Jayachamarajendra College of Engineering *Mysuru, India*
GPA: 8.47/10.0

WORK EXPERIENCE

Senior Research Scientist Jan 2025 - Sep 2026
Frontera Health *Remote / India*

- Built and productionized multimodal document-AI pipelines using Gemini vision, PDF rasterization, typed schemas, and gold-set regression tests, achieving 94–98% field accuracy across clinical assessments and expanding support to ABLLS, AFLS, ADI-R, CARS-2, Intake, and RIAS-2.
- Fine-tuned RF-DETR on COCO-formatted assessment grids and integrated it into an ABLLS vision parser, reaching 0.772 AP and 0.936 AP50 on an unseen test split versus 0.583 AP and 0.846 AP50 for the strongest SSD baseline.
- Designed an end-to-end speaker-signature pipeline using Silero VAD, pyannote diarization, WeSpeaker embeddings, similarity-graph clustering, and human review, enabling learner/therapist voice registration and achieving 80.2% segment accuracy and 92.2% mean person-level accuracy in clinic evaluations.
- Productionized a multimodal scream/cry detection pipeline combining HumanOmni video-audio inference, YAMNet filtering, and Gemini validation; evaluated 59.1 hours across 65 therapy videos and contributed to a documented test precision/F1 of 0.785/0.751.
- Led evaluation of multimodal and pose-based approaches for rare maladaptive behaviors in long therapy videos, comparing VideoMamba, YAMNet, HumanOmni, and RGB+pose pipelines; identified a candidate cascade that removed 91.5% of normal chunks while retaining 73% of positives.

Senior Computer Vision Engineer Dec 2023 - Dec 2024
Frontera Health *Remote / India*

- Contributed to the production social-engagement computer-vision stack spanning person tracking, child/adult detection, gaze models, and Gemini-based engagement summarization, including early pipeline integration, visualization tooling, and cross-model evaluation on clinic video.
- Improved social-gaze estimation in a therapy-video engagement prototype by fusing gaze-model outputs with YOLOv4 head detection and WHENet head-pose estimation, using rule-based correction to reduce false child–therapist look-at predictions.
- Improved clinical-video ASR by resolving ghost transcripts through speech-segment language voting and more reliable language detection.

Senior Engineer, Data Science Mar 2023 - Dec 2023
Colt Technology Services *Bengaluru, India*

- Implemented and integrated NLP models for contract data mining, sentiment analysis, and opportunity prioritization.
- Evaluated scalable NLP techniques and ran biweekly ML/NLP sessions to drive adoption across stakeholders.

Head Machine Learning and Technology Architect July 2022 - Feb 2023
CogniAble *Remote, India*

- Researched and implemented a few computer vision approaches for clinical screening.
- Implemented a neural network-based recommender system for treatment personalization and skill enhancement for patients with neurodevelopmental disorders

Machine Learning Engineer - Computer Vision Mar 2021 - July 2022
CogniAble *Remote, India*

- Trained and deployed a custom action-recognition network for children's behaviors and led nine production computer-vision models.
- Built object classifiers/trackers for clinical skill recognition and large-scale video training/inference pipelines.
- Implemented treatment-personalization classifiers spanning classical models, recommenders, and graph methods.

Engineer June 2018 - Mar 2021
L&T Technology Services *Bengaluru, India*

- Developed three computer-vision edge-inference applications for a leading AI/semiconductor customer, covering detection, tracking, and on-device deployment.
- Designed an OCR pipeline for HMI screen automation (2× time savings) and accelerated customer deep models with OpenVINO.
- Proposed CV framework improvements, taught deep learning for computer vision, and automated delivery workflows.

PUBLICATIONS

- "Computer Vision-Based Assessment of Autistic Children...", **Varun Ganjigunte Prakash** et al., *IEEE Access, 2023*
- "Deep Learning-Based Human Action Recognition Framework to Assess Children on the Risk of Autism...", Manu Kohli, Arpan Kumar Kar, **Varun Ganjigunte Prakash**, Prathosh A. Prasad, *ICONIP, Springer, 2023*
- "Video-based real-time assessment and diagnosis of autism spectrum disorder using deep neural networks", **Varun Ganjigunte Prakash** et al., *Expert Systems, Wiley, 2023*
- "Behavior-Aware Robot Navigation with Deep Reinforcement Learning", **Varun Ganjigunte Prakash**, *IEEE CSITSS, 2022*
- "Autonomous Service Robot", Arshad Javeed, **Varun Ganjigunte Prakash**, Sudarshan Patilkulkarni, *AIR 2019 – ACM ICPS*

TECHNICAL SKILLS

Languages:	Python, C++, MATLAB, C, Embedded C
Computer Vision:	RF-DETR, YOLO-World, YOLOv8, SSD, action recognition, pose / head-pose, person tracking, OpenCV
Multimodal & Speech:	Gemini Vision, HumanOmni, InternVideo2, VideoMamba, V-JEPA, YAMNet, WeSpeaker, Wav2Vec2, pyannote
ML / MLOps:	PyTorch, TensorFlow, Hugging Face, Temporal, AWS (S3, Batch), Docker, CUDA, W&B, Label Studio

ACCOLADES AND LEADERSHIP

- Reviewer for Biomedical Signal Processing and Control and International Journal of Machine Learning and Cybernetics.
- Co-led "Dexterous Service Robot," a top-seven Anveshan Fellowship 2018 finalist (Analog Devices).
- DELF B1 French certification; Best Teacher of the Year 2019–20 at U&I.